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B. Tech. III - Sem. (Main) Exam., May – 2023

Computer Science and Engineering

3CS4-04 Object Oriented Programming Using C++

Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time: 3 Hours **Maximum Marks: 70**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All **ten** questions are compulsory.*

Part – B: *Analytical/Problem solving questions 5 × 4 marks = 20 marks. Candidates have to answer **five** questions out of **seven**.*

Part – C: *Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer **three** questions out of **five**.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART – A

Q.1 Explain the essential looping constructs used of object oriented languages.

Q.2 Discuss data encapsulation in context of object oriented systems.

Q.3 Explain essential features of object oriented languages.

Q.4 Discuss essential data types implemented in object oriented programs.

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Q.5 What is a constructor and destructor? What is constructor overloading?

Q.6 What is polymorphism? Explain run time polymorphism.

Q.7 What are template functions? Explain with example.

Q.8 What is multiple inheritance? Explain ambiguity problem in multiple inheritance.

Q.9 What is 'this' pointer? Explain its significance.

Q.10 What is virtual destructor? Discuss the implementation of virtual destructor in object oriented languages.

PART – B

Q.1 Explain namespaces and operators in object oriented programming.

Q.2 Illustrate the concept of member functions and function overriding in an object oriented programming. Write an example program to explain the concept.

Q.3 What are friend functions? Write a program to illustrate operator overloading using friend functions.

Q.4 Explain the concept of abstract classes. Discuss the significance of abstract class in object oriented implementation.

Q.5 What are functions? What is function overloading? Explain with example program.

Q.6 Discuss template classes. Write a program to illustrate template class.

Q.7 Explain the essential phases of software development using object oriented methodology.

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PART – C

Q.1 What is STL in C++? Write a C++ program to implement STACK data structure using STL.

Q.2 Write a C++ program to illustrate hybrid inheritance. Also discuss the ambiguity in hybrid inheritance and give the solution to the problem.

Q.3 Discuss new and delete operators for memory management in object oriented programming.

Q.4 Differentiate between function overloading and function overriding. Give a C++ program to illustrate the concept of function overriding.

Q.5 Discuss containers, functions and iterators in STL in C++.

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